

Big Data

Chapter 1

Introduction to Big Data

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Programme / Syllabus

1. Introduction to Big Data
2. What are NoSQL databases?
3. Characteristics of NoSQL databases
4. Classification of the NoSQL databases
5. NewSQL databases
6. Benchmarks for Database Evaluation
7. Big Data - Introduction to Distributed Processing
8. Big Data - Storage
9. Big Data - Processing

Programme (1/3) - FUC

1. Introduction to Big Data
- Big data definition
- A little of history
- The first database revolution
- The second database revolution
- The third database revolution

2. What are NoSQL databases?
- What is wrong with the relational model?
- Big Data
- NoSQL

3. Characteristics of NoSQL databases
- NoSQL Architectures
- Data Schemas
- Data Distribution and Sharding
- Consistency
- ACID and BASE models

Programme (2/3) - FUC

4. Classification of the NoSQL databases
- Key-Value
- Document
- Column
- Graph

5. NewSQL Databases
- Main features
- Functionalities
- Differences between SQL, NoSQL and NewSQL

6. Benchmarks for Database Evaluation
- Relevant properties of a NoSQL benchmark
- Benchmarks for Key-value databases
- Benchmarks for Document databases
- Benchmarks for Column databases
- Benchmarks for Graph databases
- Examples of practical works

Programme (3/3) - FUC

7. Big Data – Introduction to Distributed Processing

- Limitations of traditional systems
- Features
- MapReduce and Hadoop
- Big Data platforms

8. Big Data - Storage

- Arquitetura
- HDFS - Hadoop Distributed File System

9. Big Data - Processing

- Tools and techniques
- Processing with Hadoop, Spark
- SQL on Hadoop, Hive
- Processing stream data

Objectives and skills to be developed

On successful completion of this curricular unit, the student should be able to:

1. Known and understand the principles and concepts of storage, processing and analysis of Big Data.

2. Using NoSQL and NewSQL databases

3. Evaluate the databases using benchmarks

4. Identify and apply big data storage, processing, and analysis concepts and techniques to real-world problems

5. Select and use appropriate tools for storing, processing and analysing large volumes of data (Big Data)

Assessment

1. Final theoretical and practical exam - 30%

2. Presentation (10%) and Report of NoSQL/NewSQL pratical work (20%) - 30%

3. Big Data practical work with report - 30%

4. Participation and continuous evaluation (5%+5%) - 10%

Exam admission conditions:

- Obtaining a minimum of 30% in each of the 2 practical works

Bibliograhy

- Santos, M. Y., & Costa, C. (2019). Big Data: Concepts, Warehousing, and Analytics. FCA.
- Erl, T., Khattak, W., & Buhler, P. (2013). Big Data Fundamentals: Concepts, Drivers & Techniques. Prentice Hall.
- Bahga, A., & Madisetti, V. (2016). Big Data Science & Analytics: A Hands-On Approach. VPT.
- Provost, F., & Fawcett, T. (2013). Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking. O'Reilly Media.
- Bengfort, B., & Kim, J. (2016). Data Analytics with Hadoop: An Introduction for Data Scientists. O'Reilly Media.
- White, T. (2015). Hadoop: The Definitive Guide: Storage and Analysis at Internet Scale. O'Reilly Media.
- Kumar, V. N., & Shindgikar, P. (2018). Modern Big Data Processing with Hadoop: Expert techniques for architecting end-to-end Big Data solutions to get valuable insights. Packt Publishing.
- Chodorow, K. (2013). MongoDB: The Definitive Guide: Powerful and Scalable Data Storage. O'Reilly Media.
- Luo, S. & Pan, L. (2026). Big Data Analysis: Theory and Technology. Springer.

Further reading (MsC Thesis)

- Bases de Dados NoSQL
  - https://recipp.ipp.pt/bitstream/10400.22/6375/1/DM_RicardoCardoso_2012_MEI.pdf
- Armazéns de dados em bases de dados NoSQL
  - https://recipp.ipp.pt/bitstream/10400.22/5681/1/DM_DanielPereira_2014_MEI.pdf
- Data migration between different data models of NoSQL databases
  - https://acervodigital.ufpr.br/handle/1884/49087

Further reading (Articles)

- Big Data Processing Tools: An experimental performance evaluation
  - Mário Rodrigues, Maribel Yasmina Santos, Jorge Bernardino
  - Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery, 2019, Vol. 9(2)
  - https://wires.onlinelibrary.wiley.com/doi/10.1002/widm.1297
- Carvalho, Inês, Filipe Sá, and Jorge Bernardino. 2023. "Performance Evaluation of NoSQL Document Databases: Couchbase, CouchDB, and MongoDB" Algorithms 16, no. 2: 78. https://doi.org/10.3390/a16020078
- Pina, Eduardo, Filipe Sá, and Jorge Bernardino. 2023. "NewSQL Databases Assessment: CockroachDB, MariaDB Xpand, and VoltDB" Future Internet 15, no. 1: 10. https://doi.org/10.3390/fi15010010

Big Data Company



Practical work on NoSQL / NewSQL

DB-Engines Ranking

- https://db-engines.com/en/ranking

429 systems in ranking, February 2026

Rank			DBMS	Database Model	Score		
Feb 2026	Jan 2026	Feb 2025			Feb 2026	Jan 2026	Feb 2025
1.	1.	1.	Oracle	Relational, Multi-model	1203.51	-33.82	-51.31
2.	2.	2.	MySQL	Relational, Multi-model	868.22	+0.70	-131.77
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model	708.14	+1.88	-78.73
4.	4.	4.	PostgreSQL	Relational, Multi-model	672.03	+5.77	+12.42
5.	5.	5.	MongoDB	Multi-model	378.73	+1.99	-17.90
6.	6.	↑ 7.	Snowflake	Relational	208.14	+0.34	+52.56
7.	7.	↓ 6.	Redis	Key-value, Multi-model	147.04	+2.88	-10.87
8.	8.	↑ 13.	Databricks	Multi-model	144.51	+2.97	+54.48
9.	9.	9.	IBM Db2	Relational, Multi-model	111.22	-1.50	-14.21
10.	10.	↓ 8.	Elasticsearch	Multi-model	106.46	-0.69	-28.17
11.	11.	11.	Apache Cassandra	Wide column, Multi-model	101.60	+0.76	-0.98
12.	12.	↓ 10.	SQLite	Relational	99.19	-1.42	-14.63
13.	13.	↑ 14.	MariaDB	Relational, Multi-model	86.09	-1.64	-3.42
14.	↑ 15.	↑ 17.	Microsoft Azure SQL Database	Relational, Multi-model	73.66	-0.53	+0.90
15.	↓ 14.	15.	Splunk	Search engine	73.05	-2.25	-7.51
16.	16.	↑ 18.	Apache Hive	Relational	73.00	-0.48	+10.51
17.	17.	↓ 12.	Microsoft Access	Relational	69.47	-2.18	-27.07
18.	18.	↓ 16.	Amazon DynamoDB	Multi-model	67.87	-1.92	-7.71
19.	19.	19.	Google BigQuery	Relational	59.58	-4.02	+4.69
20.	20.	20.	Neo4j	Graph	49.26	-0.89	+3.92
21.	21.	↑ 24.	Apache Solr	Search engine, Multi-model	33.26	-0.84	+0.58
22.	22.	↑ 23.	Teradata	Relational, Multi-model	32.44	-1.25	-1.44
23.	23.	↓ 22.	SAP HANA	Relational, Multi-model	31.75	-0.72	-3.47
24.	24.	↓ 21.	FileMaker	Relational	30.19	-1.04	-10.03
25.	25.	25.	SAP Adaptive Server	Relational, Multi-model	26.18	-0.60	-2.44
26.	26.	↑ 27.	Microsoft Azure Cosmos DB	Multi-model	23.88	-0.92	+1.75

DB-Engines Ranking: Key value

https://db-engines.com/en/ranking/key-value+store

Redis

Memcached

Etcd

Hazelcast

Ehcache

Riak KV

DB-Engines Ranking: Column

https://db-engines.com/en/ranking/wide+column+store

Cassandra

HBase

ScyllaDB

Accumulo

SWC-DB

ClickHouse

Elassandra

DB-Engines Ranking: Document

https://db-engines.com/en/ranking/document+store

MongoDB

Couchbase

CouchDB

Realm

RavenDB

RethinkDB

PouchDB

DB-Engines Ranking: Graph

https://db-engines.com/en/ranking/graph+dbms

Neo4J

Memgraph

JanusGraph

ArangoDB

OrientDB

NebulaGraph

DB-Engines - NewSQL

CockroachDB

YugaByteDB

TiDB

VoltDB

Singlestore

Vitesse

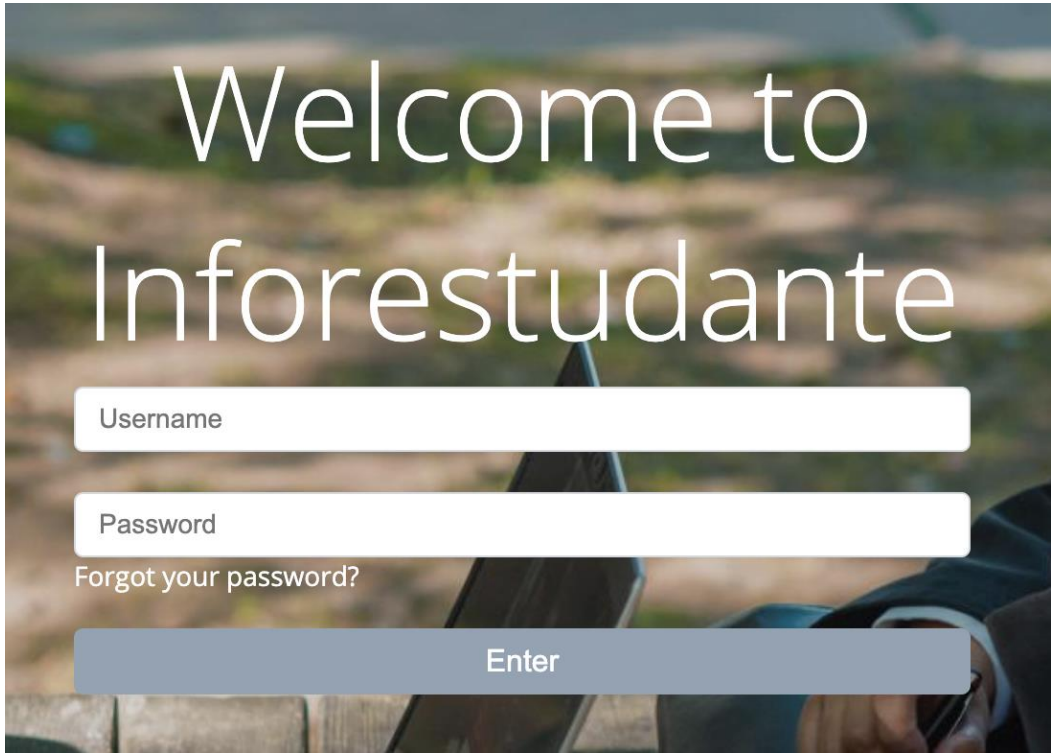
1st Practical work (Part 1) – Presentations 15-20min

- 25 February : Key-value
- 4 March : Column
- 11 March : Document
- 18 March : Graph
- 25 March : NewSQL

1st Practical work (**Part 2**) - Report

Report with experimental evaluation.

Deadline: **8 April at 6pm**



Practical work

Phase 1: Presentation of the main features, functionalities, and installation of the NoSQL/NewSQL database

Phase 2: Delivery of a detailed report on the study and evaluation of the NoSQL/NewSQL database. This report should contain a description of the installation and experimental evaluation of the respective NoSQL/NewSQL database.

This report should be written in Word or compatible format, according to the template provided, and the following structure is suggested:

Practical work - Report

Title: title of the report

Abstract: Summary of the work, not exceeding 200 words.

Introduction: Introduction to the topic of NoSQL/NewSQL databases. The importance of databases and the relevance of the type of NoSQL/NewSQL database being studied should be described. This should be followed by a summary description of the essential points and contributions of this work. Finally, the structure of the report should be described.

State of the Art: Description of other work that has been done on this topic. Other NoSQL/NewSQL databases of the same type may also be mentioned.

Body of the Work (one or more sections): This section should describe all the work carried out.

Conclusions: Main conclusions of the work carried out.

References: List of references and citations in the text using APA style. References should be listed in alphabetical order.

Appendix: Description of the steps required to install the NoSQL/NewSQL database.

Questions ??



Distribution of practical work





Agenda

- Big data definition
- A little of history
- The first database revolution
- The second database revolution
- The third database revolution
- Conclusions


HBR.ORG
Harvard Business Review
GETTING CONTROL OF BIG DATA
How vast new streams of information are changing the art of management
PAGE 59


Harvard Business Review
ANALYTICS
Data Scientist: The Sexiest Job of the 21st Century
by Thomas H. Davenport and D.J. Patil

Big Data (Definitions)

- The term “big data” refers to data that is so large, fast or complex that it's difficult or impossible to process using traditional methods.

- The act of accessing and storing large amounts of information for analytics has been around a long time.

Big Data (Wikipedia)

- Big data primarily refers to data sets that are too large or complex to be dealt with by traditional data-processing application software. Data with many entries (rows) offer greater statistical power, while data with higher complexity (more attributes or columns) may lead to a higher false discovery rate.

- Though used sometimes loosely partly due to a lack of formal definition, the best interpretation is that it is a large body of information that cannot be comprehended when used in small amounts only.

Big Data (Wikipedia)

- The term Big Data has been in use since the 1990s, with some giving credit to **John Mashey** for popularizing the term. Big data usually includes data sets with sizes beyond the ability of commonly used software tools to capture, curate, manage, and process data within a tolerable elapsed time.
  Big data philosophy encompasses unstructured, semi-structured and structured data, however the main focus is on **unstructured data**. Big data "size" is a constantly moving target; as of 2012 ranging from a few dozen terabytes (10^{12}) to many zettabytes (10^{21}) of data.

- Big data requires a set of techniques and technologies with new forms of integration to reveal insights from datasets that are diverse, complex, and of a massive scale.

John Mashey



Mashey in 2024

Born	John R. Mashey 1946 (age 79–80)
Alma mater	Pennsylvania State University (PhD)
Spouse	Angela Hey

- **John R. Mashey** (born 1946) is an American computer scientist, director and entrepreneur. He is a consultant for Techviser, a boutique consulting firm.

https://en.wikipedia.org/wiki/John_Mashey

Big Data (Wikipedia)

- A 2018 definition states:
- "Big data is where parallel computing tools are needed to handle data", and notes,
- "This represents a distinct and clearly defined change in the computer science used, via parallel programming theories, and losses of some of the guarantees and capabilities made by relational model".

Spotlight

ARTWORK Tamar Cohen, *Andrew J Buboltz*
2011, silk screen on a page from a high school yearbook, 8.5" x 12"

Data Scientist:

The Sexiest Job of the 21st Century

Meet the people who can coax treasure out of messy, unstructured data.

by Thomas H. Davenport and D.J. Patil



Harvard Business Review, October 2012


4 September 2008 | www.nature.com/nature
THE INTERNATIONAL WEEKLY JOURNAL OF SCIENCE
nature
THE BITER BIT
Viral infections for viruses
TROPICAL CYCLONES
The strong get stronger
BLACK HOLE PHYSICS
A new window on the Galactic Centre
BIG DATA
NATUREJOBS
Minnesota musings
SCIENCE IN THE PETABYTE ERA


Northeastern University
Highest-Paying Big Data Careers
1 Big Data Engineer
Median Salary: $151,300
2 Data Architect
Median Salary: $137,000
3 Data Modeler
Median Salary: $130,800
4 Data Scientist
Median Salary: $122,100
5 Database Developer
Median Salary: $109,300
6 Database Manager
Median Salary: $106,400
7 Database Administrator
Median Salary: $105,300
8 Database Security Analyst
Median Salary: $97,500
9 Business Intelligence Analysts
Median Salary: $97,500
10 Database Analyst
Median Salary: $92,900
Source: Lightcast™ Analyst, 2022

Dan Ariely

- Dan Ariely is an Israeli-American professor and author.
- He serves as a James B. Duke Professor of psychology and behavioral economics at Duke University.

https://www.azquotes.com/quote/661939

Dan Ariely



Ariely in 2019

Born	April 29, 1967 (age 58) New York City, U.S.
Education	Cognitive psychology (PhD) Business administration (PhD)
Alma mater	Duke University University of North Carolina at Chapel Hill Tel Aviv University
Known for	Behavioral economics
	Scientific career
Fields	Behavioral economics Decision making
Institutions	Duke University Massachusetts Institute of Technology
Doctoral advisor	James Bettman John G. Lynch Jr.
Website	danariely.com

A little of history

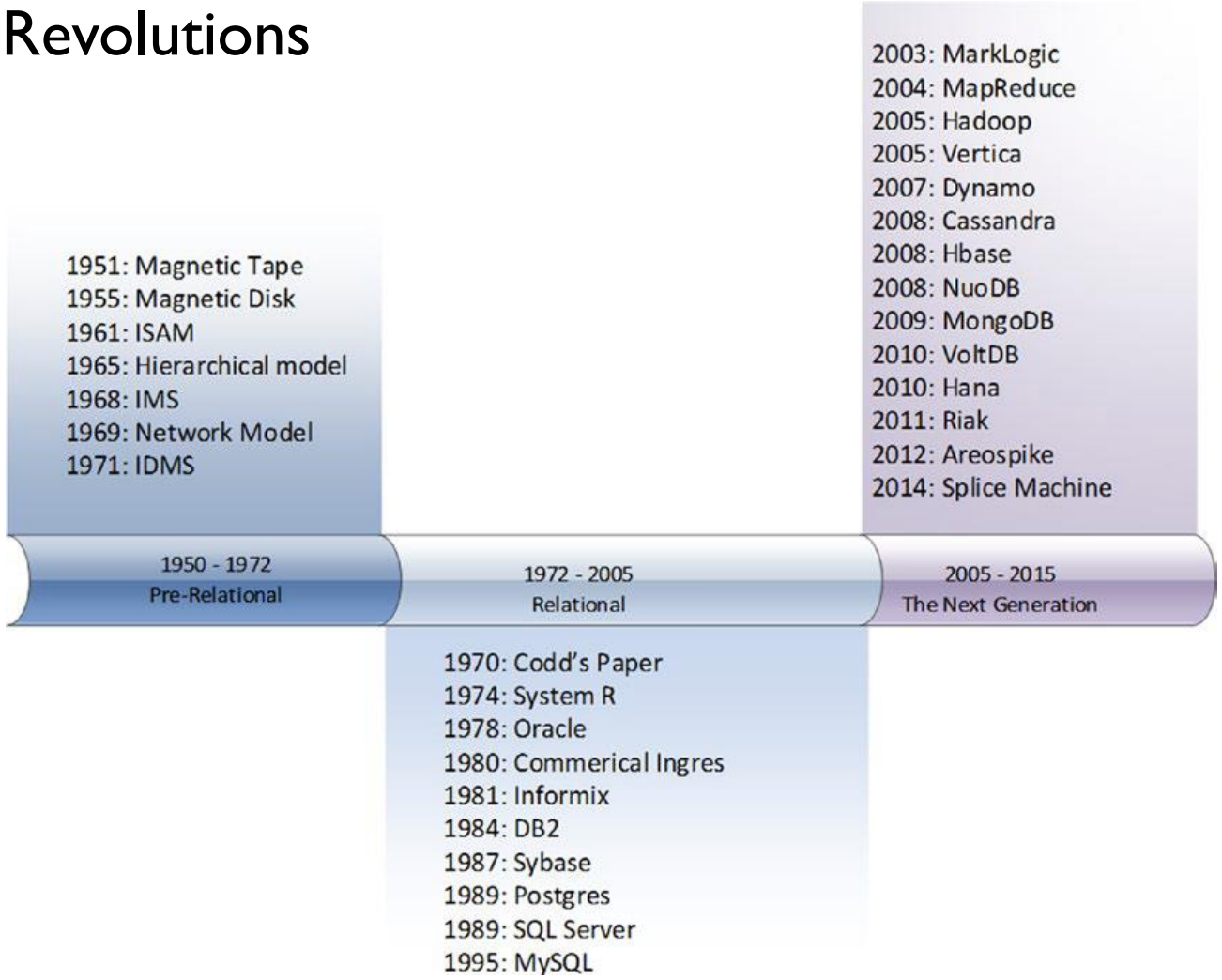
- Three Database Revolutions

What was the historic milestone in Databases?

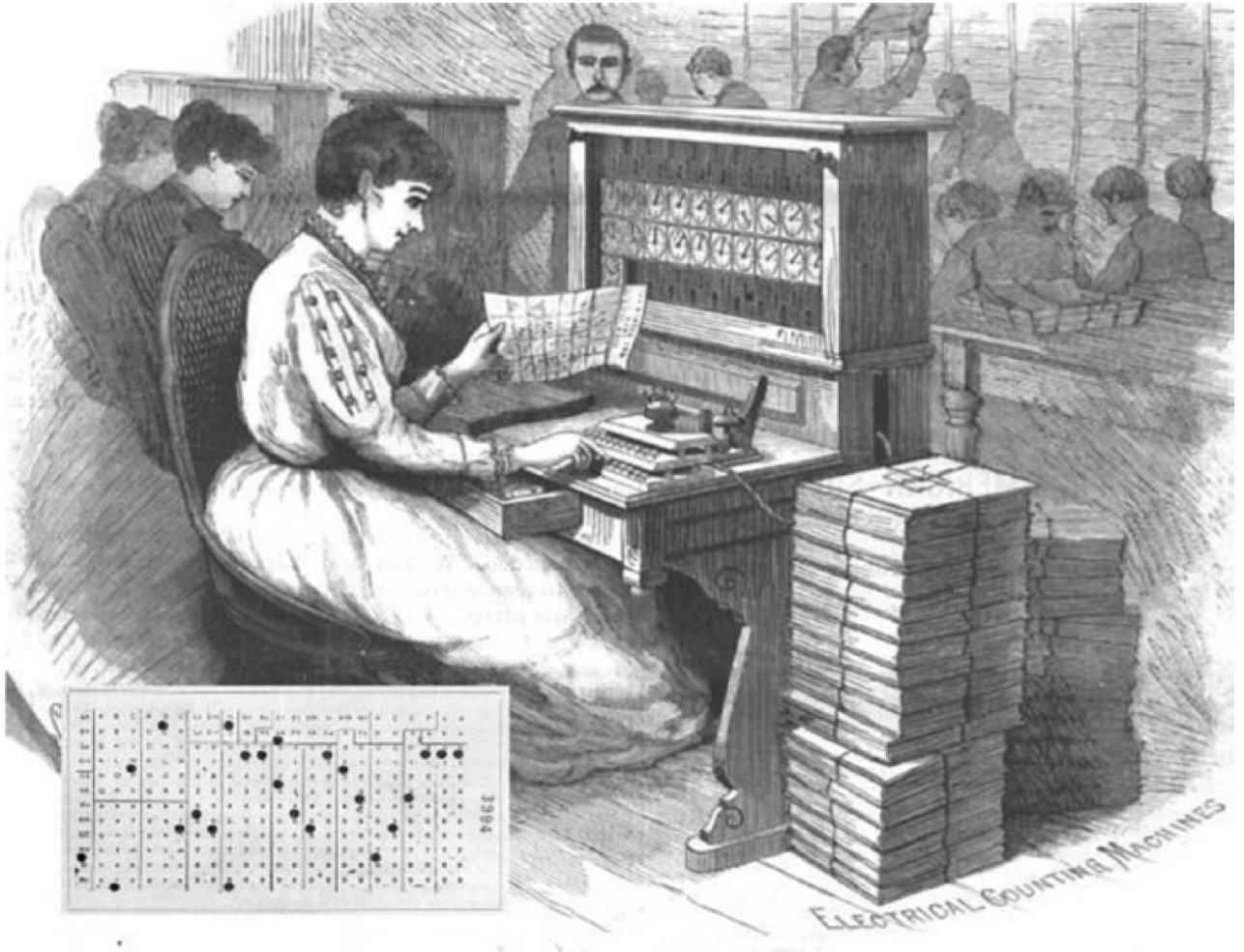


A little of history

- Three Database Revolutions



Early Database Systems



Tabulating machines and punched cards used to process 1890 U.S. census

A little of history

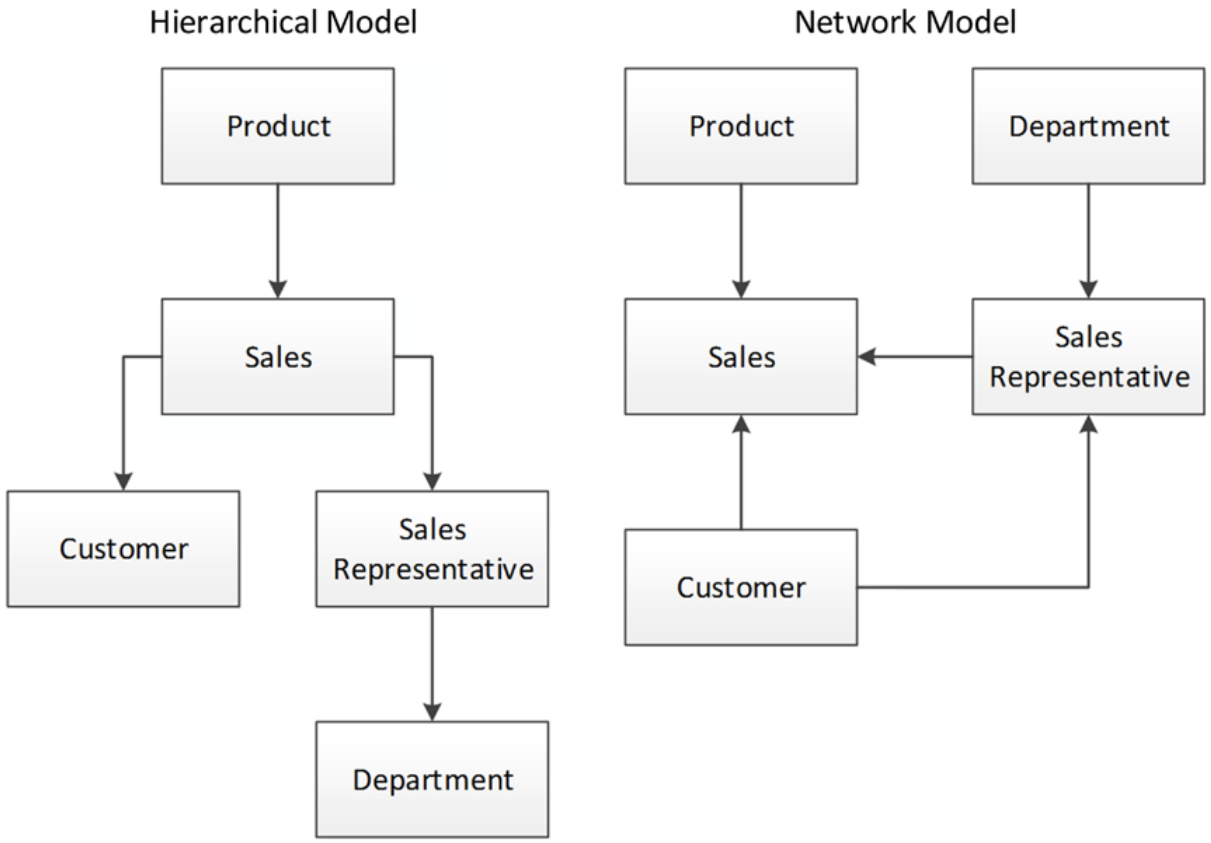
- The emergence of electronic computers following the Second World War represented the first revolution in databases.
- Early "databases" used paper tape initially and eventually magnetic tape to store data sequentially.
- The development of indexing methods such as ISAM (Index Sequential Access Method) made fast record-oriented access feasible and consequently allowed for the birth of the first OLTP (On-line Transaction Processing) computer systems.
- ISAM and similar indexing structures powered the first electronic databases. However, these were completely under the control of the application - there were databases but no Database Management Systems (DBMS).

The First Database Revolution

- Requiring every application to write its own data handling code was clearly a productivity issue
  - Every application had to reinvent the database wheel
- Therefore, it became desirable to externalize database handling logic from the application in a separate code base.
  - This layer—the DataBase Management System, or DBMS—would minimize programmer overhead and ensure the performance and integrity of data access routines
- These first-generation databases ran exclusively on the mainframe computer systems of the day — largely IBM mainframes.

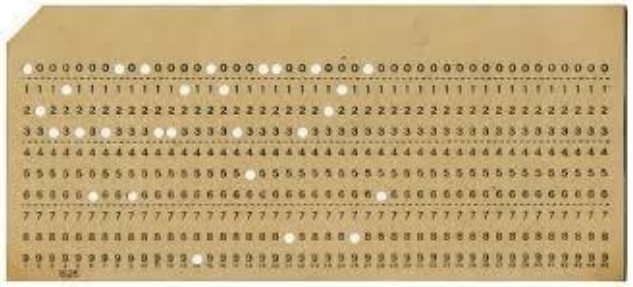
The First Database Revolution

- By the early 1970s, two major models of DBMS were competing : ***hierarchical model*** and the ***network model***



The Second Database Revolution

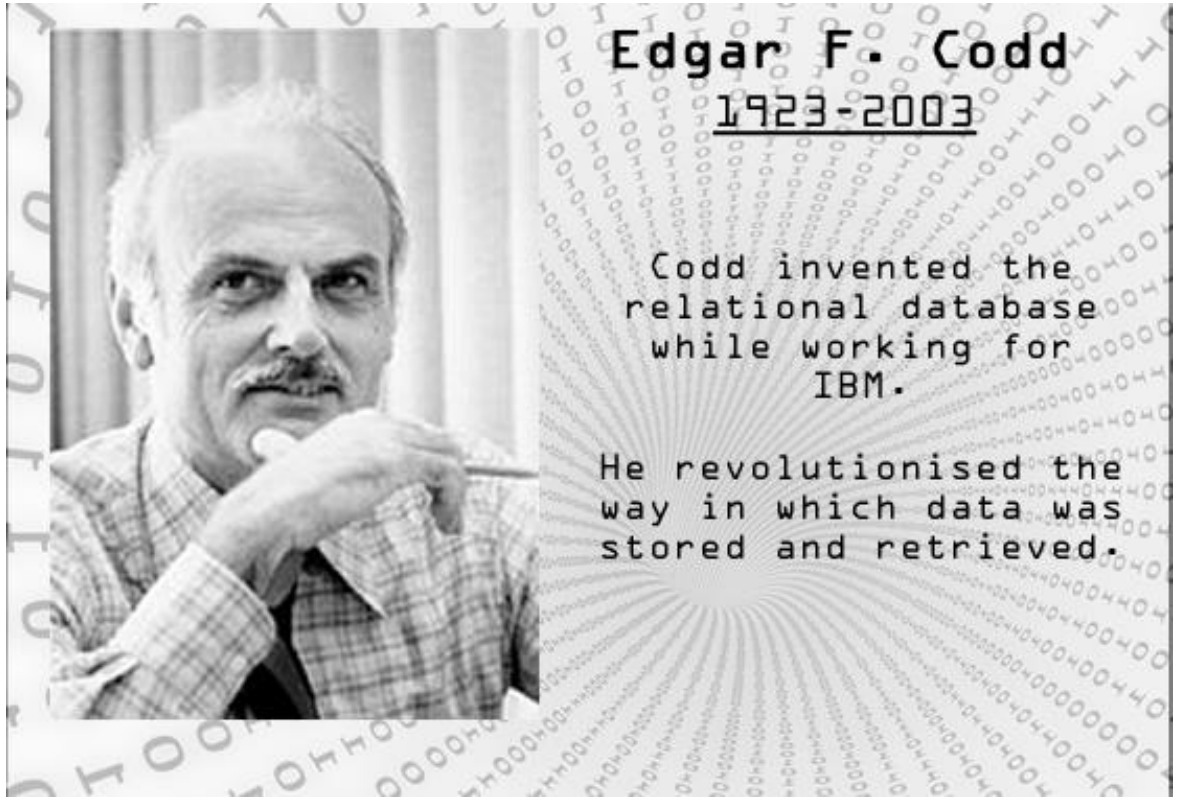
Who was the author of the relational model?





The Second Database Revolution

- No single person has had more influence over database technology than **Edgar Codd**



The Second Database Revolution

- Codd was very familiar with the databases of the day, and he pointed significant reservations about their design:
  - **Existing databases were too hard to use**. Databases of the day could only be accessed by people with specialized programming skills.
  - **Existing databases lacked a theoretical foundation**. Codd's mathematical background encouraged him to think about data in terms of formal structures and logical operations.
  - **Existing databases mixed logical and physical implementations**. The representation of data in existing databases matched the format of the physical storage in the database, rather than a logical representation of the data that could be comprehended by a nontechnical user.

The Second Database Revolution

- Codd published his 1970 paper "A Relational Model of Data for Large Shared Data Banks", *Communications ACM*
  - This classic paper contained the core ideas that defined the relational database model that became the most significant—almost universal—model for database systems for a generation
- Key concepts of the relational model:

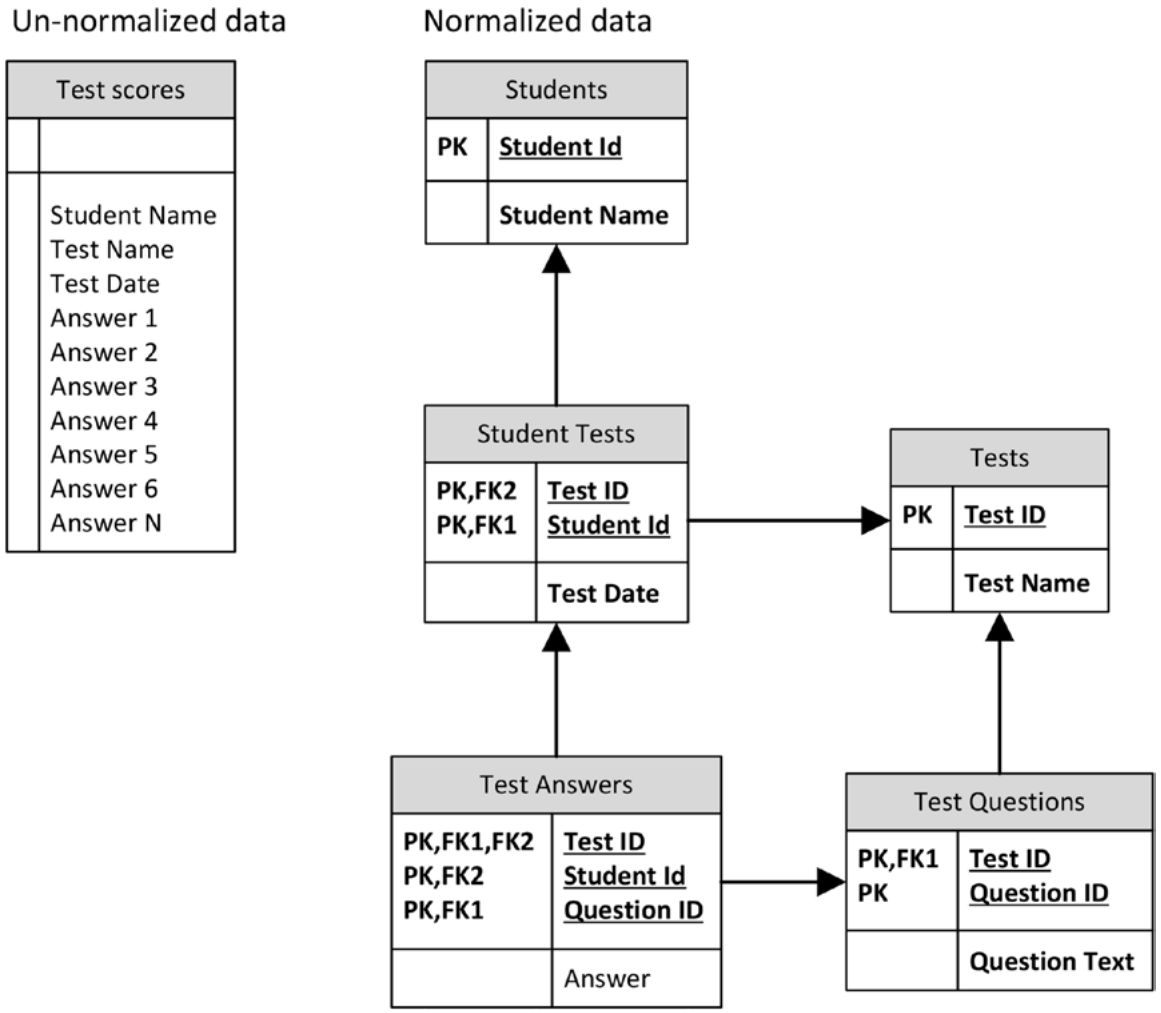
  • Tuples, an unordered set of attribute values. In an actual database system, a tuple corresponds to a row, and an attribute to a column value.

  • A relation, which is a collection of distinct tuples and corresponds to a table in relational database implementations.

  • Constraints, which enforce consistency of the database. Key constraints are used to identify tuples and relationships between tuples.

  • Operations on relations such as joins, projections, unions, and so on.

The Second Database Revolution



Normalized and un-normalized data

The Second Database Revolution

Transaction Models

- Jim Gray defined the most widely accepted transaction model in the late 1970s : "*A transaction is a transformation of state which has the properties of atomicity (all or nothing), durability (effects survive failures) and consistency (a correct transformation).*"
- This became popularized as ACID transactions:
- Atomic, Consistent, Isolated, and Durable. An ACID transaction should be:
  - Atomic: The transaction is indivisible—either all the statements in the transaction are applied to the database or none are.
  - Consistent: The database remains in a consistent state before and after transaction execution.
  - Isolated: While multiple transactions can be executed by one or more users simultaneously, one transaction should not see the effects of other in-progress transactions.
  - Durable: Once a transaction is saved to the database (in SQL databases via the COMMIT command), its changes are expected to persist even if there is a failure of operating system or hardware.
- The restriction on scalability beyond a single data center implied by the ACID transaction model has been a key motivator for the development of new database architectures.

The Second Database Revolution

The First Relational Databases

- Initial reaction to the relational model was somewhat unenthusiastic. Existing vendors, including IBM, had sincere reservations about the ability of a system to deliver adequate performance if the data representation was not fine-tuned to the underlying access mechanisms.
- IBM did, however, initiate a research program to develop a prototype relational database system in 1974, called System R.
  - System R demonstrated that relational databases could deliver adequate performance, and it pioneered the SQL language. (Codd had specified that the relational system should include a query language but had not mandated a specific syntax)
- Also, during this period, Mike Stonebraker at Berkeley started work on a database system that was called INGRES.
  - INGRES was also relational, but it used a non-SQL query language called QUEL.
- At this point, Larry Ellison enters the story. Ellison was more entrepreneurial than academic by nature, though extremely technically sophisticated, having worked at Amdahl. Ellison was familiar both with Codd's work and with System R, and he believed that relational databases represented the future of database technology.
- In 1977, Ellison founded the company that would become Oracle Corporation and which would release the first commercially successful relational database system.

The Second Database Revolution

Database Wars !

- By 1981, IBM had released a commercial relational database called SQL/DS, but since it only ran on IBM mainframe operating systems, it had no influence in the rapidly growing minicomputer market.
- Ellison's Oracle database system was commercially released in 1979 and rapidly gained traction on the minicomputers provided by companies such as Digital and Data General.
- Relational databases became so dominant that the vendors of the older database systems became obliged to describe their offerings as also being relational. This prompted Codd to compose his famous 12 rules:
  - https://www.red-gate.com/simple-talk/databases/theory-and-design/codds-twelve-rules/
- During the succeeding decades many new database systems were introduced. These include Sybase, Microsoft SQL Server, Informix, MySQL, and DB2.
- While each of these systems attempts to differentiate by claiming superior performance, availability, functionality, or economy, they are virtually identical in their confidence on three key principles: Codd's relational model, the SQL language, and the ACID transaction model.

The Second Database Revolution

- When we say RDBMS, we generally refer to a database that implements the relational data model, supports ACID transactions, and uses SQL for query and data manipulation

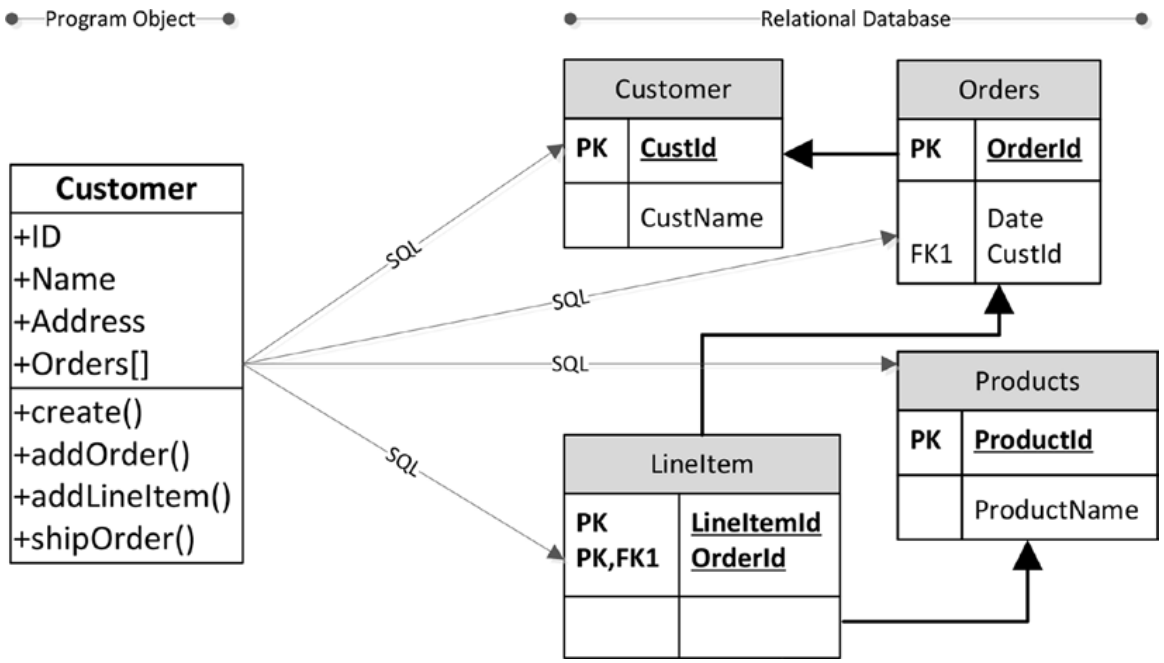
- Client-server Computing
  - Client-server allowed for a richness of experience that was unparalleled, and by the early '90s, virtually all new applications aspired to the client-server architecture.

- Object-oriented Programming and the OODBMS
  - Object-oriented (OO) programming merged attributes and behaviours into a single object. So, for instance, an employee object might represent the structure of employee records as well as operations that can be performed on those records—changing salary, promoting, retiring, and so on.
    - **Encapsulation**: An object class encapsulates both data and actions (methods) that may be performed on that data
    - **Inheritance**: Object classes can inherit the characteristics of a parent class.

The Second Database Revolution

OODBMS

- When an object was stored into or retrieved from a relational database, multiple SQL operations would be required to convert from the Object-Oriented representation to the relational representation. This could lead to performance or reliability issues.



The Second Database Revolution

OODBMS

- Object Oriented Database Management System (OODBMS) would store program objects directly without normalization and would allow applications to load and store their objects easily.
- The object-oriented database movement created a manifesto outlining the key arguments for and properties of OODBMS:

  https://www.cs.cmu.edu/~clamen/OODBMS/Manifesto/
- In implementation, OODBMS resembles the navigational model from the pre-relational era - pointers within one object (a customer, for instance) would allow navigation to related objects (such as orders).
- However, by the end of the decade, OODBMS systems had completely failed to gain market share. Mainstream database vendors such as Oracle and Informix had successfully implemented many OODBMS features, but even these features were rarely used.

The Second Database Revolution

The Relational Plateau

- Once the excitement over object-oriented databases had run its course, relational databases remained unchallenged until the latter half of the 2000s.
- In fact, for a period of roughly 10 years (1995–2005), no significant new databases were introduced: there were already enough RDBMS systems to saturate the market.
- Considering that this period essentially represents a time when the Internet grew from geeky curiosity to an all-pervasive global network, that no new database architectures emerged during this period is astonishing, and it is a testament to the power of the relational model.

The Third Database Revolution

- The difference in application architectures between the client-server era and the era of massive web-scale applications created pressures on the relational database that could not be improved through incremental innovation.

Google and Hadoop

- The relational database was inadequate to deal with the volumes and velocity of the data confronting Google.
- The challenges that enterprises face with "big data" today are problems that Google first encountered almost 20 years ago.
- Very early on, Google had to invent new hardware and software architectures to store and process the exponentially growing quantity of websites it needed to index.

The Third Database Revolution

Google and Hadoop

- In 2003, Google revealed details of the distributed file system GFS that formed a foundation for its storage architecture, and in 2004 it revealed details of the distributed parallel processing algorithm **MapReduce**, which was used to create World Wide Web indexes.
- In 2006, Google revealed details about its **BigTable** distributed structured database.
- These concepts, together with other technologies, many of which also came from Google, formed the basis for the **Hadoop** project, which matured within Yahoo! and which experienced rapid acceptance from 2007 on.
- The Hadoop ecosystem more than anything else became a technology enabler for the Big Data ecosystem.

The Third Database Revolution

The Rest of the Web

- While Google had an overall scale of operation and data volume way beyond that of any other web company, other websites had challenges of their own. Websites dedicated to online e-commerce - Amazon, for example - had a need for a transactional processing capability that could operate at massive scale.
- Many early websites attempted to scale open-source databases through a variety of do-it-yourself techniques. Sharding involves partitioning the data across multiple databases based on a key attribute, such as the customer identifier.
- Sharding at sites like Facebook has allowed a MySQL-based system to scale up to massive levels, but the downsides of doing this are immense. Many relational operations and database-level ACID transactions are lost. It becomes impossible to perform joins or maintain transactional integrity across shards.
- Meanwhile, a similar dilemma within Amazon had resulted in development of an alternative model to strict ACID consistency within its source data store. Amazon revealed details of this system, "Dynamo," in 2008.
- Amazon's Dynamo model, together with innovations from web developers looking for a "webscale" database, led to the emergence of what came to be known as **key-value databases**.

The Third Database Revolution

Cloud Computing

- The existence of applications and databases "in the cloud" - that is, accessed from the Internet - had been a persistent feature of the application landscape since the late 1990s. However, around 2008, cloud computing erupted somewhat abruptly as a major concern for large organizations and a huge opportunity for startups.
- Between 2006 and 2008, Amazon rolled out Elastic Compute Cloud (EC2). EC2 made available virtual machine images hosted on Amazon's hardware infrastructure and accessible via the Internet.
- Amazon added other services such as storage (S3, EBS), Virtual Private Cloud (VPC), a MapReduce service (EMR), and so on.
- The entire platform was known as Amazon Web Services (AWS) and was the first practical implementation of an Infrastructure as a Service (IaaS) cloud.
- AWS became the inspiration for cloud computing offerings from Google, Microsoft, and others.

The Third Database Revolution

Document Databases

- Starting about 2004, an increasing number of websites were able to offer a far richer interactive experience than had been the case previously.
- This was enabled by the programming style known as **AJAX** (Asynchronous JavaScript and XML), in which JavaScript within the browser communicates directly with a backend by transferring XML messages.
- XML was soon superseded by **JSON** (JavaScript Object Notation), which is a self-describing format similar to XML but is more compact and tightly integrated into the JavaScript language.
- JSON became the *de facto* standard for storing objects to disk.
- Some websites started storing JSON documents directly into columns within relational tables.
- Therefore, people decided to eliminate the relational middleman and create a database in which JSON could be directly stored. These became known as **document databases**.

The Third Database Revolution

The "NewSQL"

- In 2007, Michael Stonebraker, pioneer of the Ingres and Postgres database systems, led a research team that published the seminal paper "The End of an Architectural Era (It's Time for a Complete Rewrite)." *In Proceedings of VLDB conference*

  http://nms.csail.mit.edu/~stavros/pubs/hstore.pdf

- This paper pointed out that the hardware assumptions that underlie the consensus relational architecture no longer applied, and that the variety of modern database workloads suggested a single architecture might not be optimal across all workloads.

The Third Database Revolution

The "NewSQL"

- Stonebraker and his team proposed a number of variants on the existing RDBMS design, each of which was optimized for a specific application workload.
- Two of these designs became particularly significant:
  - **H-Store** described a pure in memory distributed database
  - **C-Store** specified a design for a columnar database.
- Both these designs were extremely influential in the years to come and are the first examples of what came to be known as **NewSQL database systems** - databases that retain key characteristics of the RDBMS but that diverge from the common architecture exhibited by traditional systems such as Oracle and SQL Server.

The Third Database Revolution

The Nonrelational Explosion

- A huge number of non-relational database systems emerged in the first half of the 2000s.
- In particular, in the years 2008–2009: MongoDB, Cassandra, and HBase.
- At first, these new breeds of database systems lacked a common name: "Distributed Non-Relational Database Management System" (DNRDBMS).
- However, in late 2009, the term NoSQL quickly became shorthand for any database system that broke with the traditional SQL database.

The Third Database Revolution

The Nonrelational Explosion

- By 2011, the term NewSQL became popularized as a means of describing this new breed of databases that, while not representing a complete break with the relational model, enhanced or significantly modified the fundamental principles - and this included columnar databases.
- Finally, the term Big Data burst onto mainstream consciousness in early 2012. Although the term refers mostly to the new ways in which data is being leveraged to create value, we generally understand "Big Data solutions" as convenient shorthand for technologies that support large and unstructured datasets such as Hadoop.

The Third Database Revolution

- NoSQL, NewSQL, and Big Data are in many respects vaguely defined, advertised, and overloaded terms. However, they represent the most widely understood phrases for referring to next-generation database technologies.
- NoSQL databases reject the constraints of the relational model, including strict consistency and schemas.
- NewSQL databases retain many features of the relational model but change the underlying technology in significant ways.
- Big Data systems are generally oriented around technologies within the Hadoop ecosystem, increasingly including Spark.

Conclusion: One Size Doesn't Fit All



Conclusion

The first revolution

- The first database revolution arose as an inevitable consequence of the emergence of electronic digital computers.
- In some respect, the databases of the first wave were electronic analogies of pre-computer technologies such as punched cards and tabulating machines.
- Early attempts to add a layer of structure and consistency to these databases may have improved programmer efficiency and data consistency, but they left the data locked in systems to which only programmers held the keys.

Conclusion

The second revolution

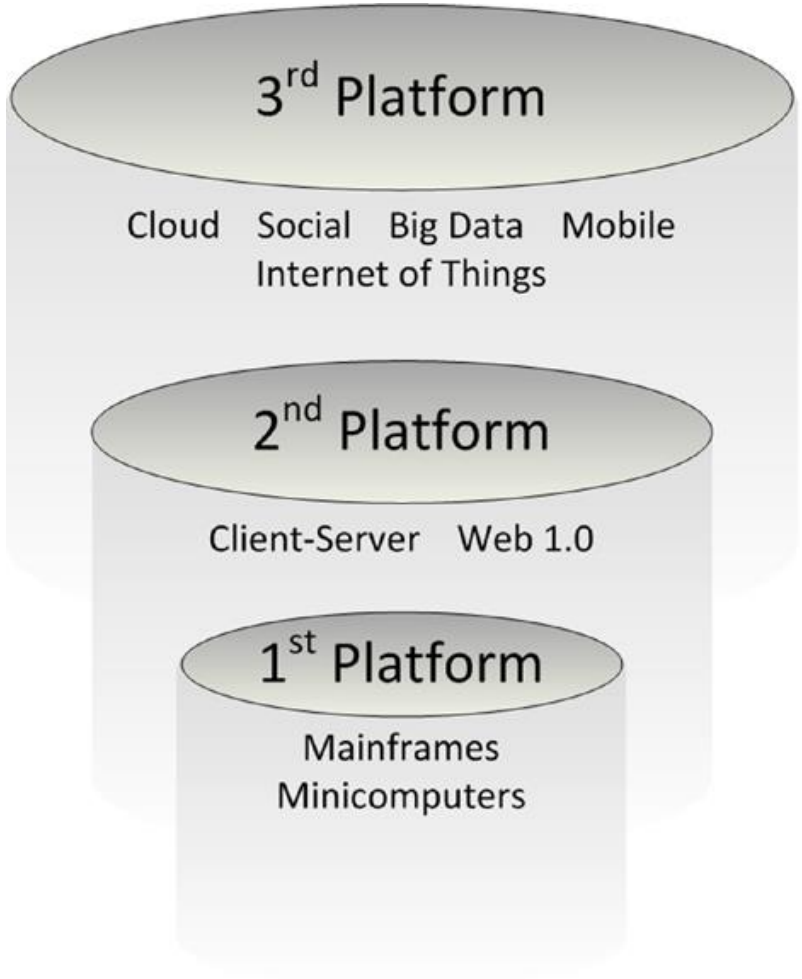
- The second database revolution resulted from Edgar Codd's realization that database systems would be well served if they were based on a solid, formal, and mathematical foundation; that the representation of data should be independent of the physical storage implementation; and that databases should support flexible query mechanisms that do not require sophisticated programming skills. The successful development of the modern relational database over such an extended time - more than 30 years of commercial dominance - represents a triumph of computer science and software engineering. Rarely has a software theoretical concept been so successfully and widely implemented as the relational database.

Conclusion

The third revolution

- The third database revolution is not based on a single architectural foundation. If anything, it rests on the proposition that a single database architecture cannot meet the challenges posed in our modern digital world.
- The existence of massive social networking applications with hundreds of millions of users and the emergence of Internet of Things (IoT) applications with potentially billions of machine inputs, overloading the relational database - and particularly the ACID transaction model - to the breaking point.
- At the other end of the scale, we have applications that must run on mobile and wearable devices with limited memory and computing power. And we are inundated with data, much of which is of unpredictable structure for which rendering to relational form is unsustainable.

Summary:
The three waves of database technology



Relational Database
NoSQL
NewSQL
Big Data platforms

Relational Database

Hierarchical Database
Network Database
ISAM files

Databases types - Tutorial

- https://www.educba.com/types-of-dbms/

Types of DBMS



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Ravi Rathore

Introduction to DBMS Types


ANY
QUESTIONS?